Unit 3 (Part 1) - The Real Number System

Definitions:

The Real number system is made up of several groups of numbers:

⇒ **Natural Numbers:**
These probably were the first numbers used. The symbols were developed long after the concept.

⇒ **Whole numbers:**
A symbol for the number 0 did not appear in any early number systems. It is thought that the Ancient Mayans were the first to use a symbol for 0, however 0 was only truly accepted as a number in the 18th Century.

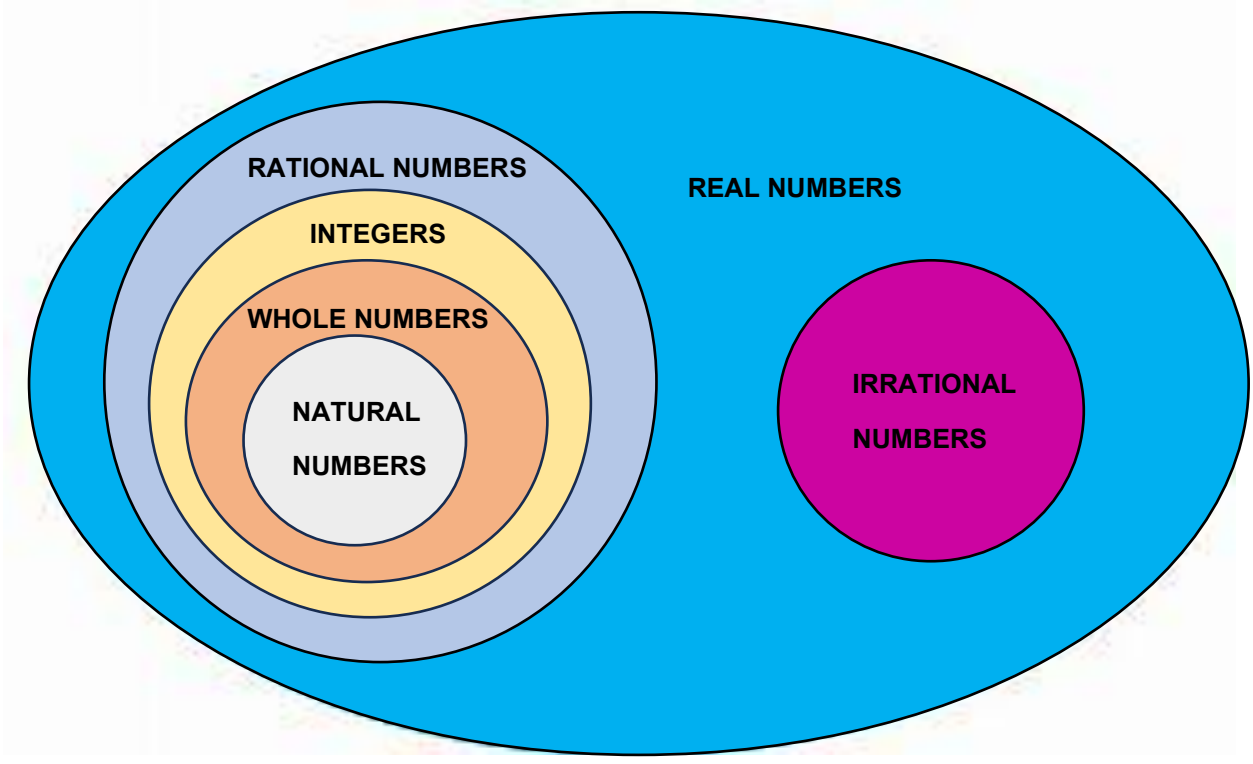
⇒ **Integers:**
The word integer comes from the word integral, meaning whole.

⇒ **Rational numbers:**
Any number that can be written as a fraction, or a perfect number.

⇒ **Irrational numbers:**
Any number that cannot be written as a fraction. The most common examples are roots, non-recurring, and, non-terminating decimals.

⇒ **Real numbers:**
The above groups combined form the group of real numbers.

⇒ **Non-real numbers:**
Numbers that cannot be classified into any of the above groups are considered to be non-real. The two most common occurrences of this are when we attempt to square root a negative number or when we divide by 0.



Summary of the Number Sets:

Number Set	Symbol	Description	Examples
Real	$\mathbb{R}$	All (1D) quantities that can exist in a real-life scenario. Every real number is either rational or irrational	
Rational	$\mathbb{Q}$	All real numbers that can be written as a quotient of integers (obviously without dividing by zero). These include all terminating and recurring decimals. A special case is a decimal value of zero.	$\left\{x \middle\| x = \frac{a}{b}\,;a,b \in \mathbb{Z}\,;b \neq 0\right\}$ -5 ; ¼ ; -2½ ; $\frac{13}{7}$ 3.271 ; $4.\overline{835}$
Irrational	$\mathbb{Q}^{\mid}$	All real numbers that cannot be expressed as a quotient of integers. These numbers have non-terminating and non-recurring decimals.	$\sqrt{2}$; $\sqrt{13}$; 4.15632 ...
Integers	$\mathbb{Z}$	All rational numbers with a decimal value of zero.	{...; -2 ; -1 ; 0 ; 1; 2 ; ...}
Whole	$\mathbb{N}_0$	The **non-negative** integers.	{0 ; 1 ; 2 ; 3 ; ...}
Natural	$\mathbb{N}$	The **positive** integers.	{1 ; 2 ; 3 ; ...}

It is important to note that all real numbers can be further classified into other sets:

- Natural numbers are contained in the whole numbers; natural numbers are a subset of the whole numbers (all natural numbers are whole numbers, but not all whole numbers are natural numbers).
- Whole numbers are contained in the integers; whole numbers are a subset of the integers (all whole numbers are integers, but not all integers are whole numbers).
- Integers are contained in the rational numbers; the integers are a subset of the rational numbers (all integers are rational, but not all rational numbers are integers).
- Real numbers are either rational or irrational.
- Complex numbers are 2-dimensional and contain a real part and an imaginary part.

Example 1: Complete the table

Name	Natural		Rational		Real
Symbol		$\mathbb{Z}$		$\mathbb{Q}'$	
8					
$\sqrt{8}$					
8,1					
$\frac{-21}{7}$					
$\sqrt{100}$					
1,111…					
-13					
$\frac{2,1}{7}$					
$\sqrt{\frac{25}{9}}$					

Homework: Ex. 1 (**A – C**) pg. 7 (Mind Action Series new edition, 2022)

<<< SELF STUDY >>>

Rounding (decimal places)

⇒ Count to the number of decimal places *after* the comma that you want to round off to.

⇒ Look at the digit to the right of this decimal place.

- If it is less than 5, "drop" it and all the digits to the right of it. This is rounding down … decimals were removed decimals, and the rounded value is smaller.
- If it is 5 or more than 5, then add one more the value of the digit immediately to the left of it and drop it and all digits to the right of it. This is rounding up … we increased the decimal value, and the rounded value is larger.

Examples

Round off the following numbers to the decimal places indicated.

a. 9,23584 (3dp) _____ _____

b. 67,236 (2dp) _____ _____

c. 4,3768534 (4dp) _____ _____

d. 17,247398 (5dp) _____ _____

e. 79,999 (3dp) _____ _____

f. 34,2784682 (4dp) _____ _____

g. 5,555555 (5dp) _____ _____

You should also be able to round off to any place value (e.g. unit, hundred, ten thousand), and also to the nearest 5, 50, 500, 5 000 etc.

Go back to the examples above and round them off to the nearest 5.